

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

Requirement-Driven Adaptive Differential Evolution for Many-Task Optimization

Zhi-Fan Tang, *Student Member, IEEE*, Zhi-Hui Zhan, *Fellow, IEEE*, Yun Li, *Fellow, IEEE*, and Jun Zhang, *Fellow, IEEE*

Abstract— The major challenge in evolutionary many-task optimization (EMaTO) is to transfer knowledge among the tasks effectively. Moving on existing EMaTO algorithms, which mainly focus on *what* kind of knowledge to select and *how* to transfer the selected knowledge, this article focuses on a novel perspective to tackle the issue of *when* to transfer knowledge. This gives rise to a requirement-driven adaptive differential evolution (RADE) algorithm. Specifically, the RADE introduces a dual-scale evolutionary state estimator (ESE) to evaluate the state of evolution from both temporal and spatial scales, obtaining the real optimization requirement of the population and thus adaptively controlling the timing of knowledge transfer. This way, RADE not only leads to efficiently dealing with the *when* issue, but also drives two adaptive strategies for knowledge selection and knowledge transfer, leading to more efficient dealing with the *what* and the *how* issues, respectively. In particular, a requirement-driven adaptive knowledge selection strategy merges the evolutionary state similarity and the population distribution similarity for enhancing the selection of relevant source knowledge. Then, a requirement-driven adaptive knowledge transfer strategy is executed based on adaptive parameters and operators, thus significantly improving the efficiency of knowledge transfer. Experimental results on the CEC19 and WCCI22 benchmarks show that RADE outperforms the compared state-of-the-art EMaTO algorithms in both effectiveness and efficiency.

Index Terms—Many-task optimization; evolutionary computation; differential evolution; evolutionary state estimator; knowledge selection; knowledge transfer

Manuscript received XXXX; revised XXXX; accepted XXXX. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFF0509600, in part by the National Natural Science Foundation of China (NSFC) under Grant 62576175 and Grant U23B2039, in part by the Tianjin Top Scientist Studio Project under Grant 24JRRRCRC00030, in part by the Tianjin Belt and Road Joint Laboratory under Grant 24PTLYHZ00250, in part by the Natural Science Foundation of Tianjin Key Program under Grant 25JCZDJC01040, and in part by the National Research Foundation of Korea (NRF) Grant funded by the Korea government (MSIT) (No.RS-2025-00555463& RS-2025-25456394). (Corresponding authors: Zhi-Hui Zhan; Jun Zhang.)

Zhi-Fan Tang is with the School of Computer Science and Engineering, South China University of Technology, Guangzhou 510006, P. R. China.

Zhi-Hui Zhan is with the College of Artificial Intelligence, Nankai University, Tianjin 300350, P. R. China and also with the School of Computer Science and Engineering, South China University of Technology, Guangzhou 510006, P. R. China (e-mail: zhanapollo@163.com).

Yun Li is with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen 518110, China, and also with the i4AI Ltd., London WCIN3AX, U.K.

Jun Zhang is with the Nankai University, Tianjin, China; Hanyang University ERICA, Ansan, South Korea; Zhejiang Normal University, Jinhua, China; Chaoyang University of Technology, Taichung, Taiwan, and Kaifeng University, Kaifeng, China.

I. INTRODUCTION

IN real-world applications, many complex optimization problems are not isolated but are intricately interconnected. While evolutionary computation (EC) [1]–[3] has shown strong capability in solving various optimization problems separately, it often fails to exploit the shared information between these problems. Drawing inspiration from transfer learning methodologies, researchers have increasingly shifted their focus toward transfer optimization, encouraging the rapid development of diverse evolutionary multi-task optimization (EMTO) algorithms [4]–[6]. These algorithms can effectively share and transfer knowledge between tasks, thereby improving the overall efficiency in optimization through the information exchange across tasks.

EMTO algorithms leverage the similarities between tasks and the commonalities in evolutionary processes to extract reusable knowledge from the experiences of other evolution and transfer the knowledge across tasks, thereby accelerating convergence speed and improving optimization performance [7][8]. EMTO algorithms have demonstrated significant success in diverse applications, such as machine learning and data science [9], computer vision [10], path planning [11], and network optimization [12].

However, as the number of tasks in a multi-tasking optimization problem increases, the complexity of inter-task interactions also rises, making it increasingly difficult to assess the validity of knowledge sharing. Especially, as the number of tasks gets more than three to become a many-task optimization problem (MaTOP) [13][14], the effectiveness of knowledge transfer decreases significantly. Consequently, a significant issue in the design of EMTO algorithms is determining how to introduce beneficial knowledge through a positive transfer strategy at a suitable time. In other words, the challenges in evolutionary many-task optimization (EMaTO) mainly manifest in three significant and interconnected issues: *what* knowledge to transfer, *how* to transfer the knowledge, and *when* to transfer the knowledge.

Firstly, to address the issue of *what* knowledge to transfer, it is essential to select an appropriate source task [15][16]. An ideal source task should have a certain degree of relevance or similarity with the target task, since this can provide valuable help for solving the target task. By selecting an appropriate source task, we can leverage its search patterns and experience to reduce the search difficulty of the target task and

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

significantly improve the optimization efficiency. Therefore, when selecting the knowledge, it is necessary to fully consider the optimization requirement of the target task to ensure the knowledge is relevant and positive, thereby maximizing the benefits.

Secondly, the issue of *how* to transfer knowledge plays a significant role in determining the effectiveness of knowledge transfer [17][18]. A well-designed transfer strategy should enable the integration of useful knowledge from the source task into the target task while avoiding negative transfer effects. The choice and implementation of knowledge transfer strategies should fully consider the optimization requirement of the target task to ensure that the transferred knowledge fits the task context, thereby improving the overall optimization efficiency.

Although certain successes and significant progresses have been made in addressing the challenges associated with *what* knowledge to transfer and *how* to implement the transfer effectively in EMaTO, the significance of determining *when* to transfer the knowledge is rarely discussed. However, deciding when to transfer knowledge is the key to making the transfer effective. The success of selecting what to transfer and how to transfer always relies on correct transfer timing. Transfer at a suitable time can improve the optimization performance. Conversely, transfer at the wrong time may disrupt a population that is already converging well, regardless of the quality of the transferred knowledge. Therefore, suitable transfer timing is essential for ensuring positive knowledge transfer.

While some researches indirectly control the transfer timing through dynamic transfer probabilities [19][20], they often lack a direct response to the dynamic optimization requirements of populations. The issue of explicitly determining the optimal timing for knowledge transfer remains largely underexplored. Therefore, a clear research gap exists for a methodology that determines transfer timing based on a direct detection of the evolutionary process.

To fill this gap, this article proposes a requirement-driven framework that directly links transfer timing to the evolutionary states of populations. The state is a dynamic property of the evolutionary process [21], characterized by the information across both in time (e.g., the convergence trend of the population with the number of generations) and in space (e.g., the solution distribution of the population in the current generation). To capture both the temporal and spatial information, we introduce a dual-scale evolutionary state estimator (ESE) that utilizes the accumulated success rate and the population diversity as indicators for the temporal and spatial scales, respectively. After detecting the optimization requirements and determining the timing of transfer by using the dual-scale ESE (deal with the *when* issue), effective knowledge transfer can be realized through an adaptive knowledge selection strategy (better deal with the *what* issue) and an adaptive knowledge transfer strategy (better deal with the *how* issue) from meeting the optimization requirement, thereby improving the optimization performance.

Based on the above discussions, we propose a requirement-driven adaptive differential evolution (RADE) algorithm. By focusing on the population's evolutionary state and tailoring the knowledge transfer process to meet the specific optimization requirement of each population, RADE can accurately identify suitable transfer timing and provide flexible and adaptable knowledge transfer during the optimization process.

The major contributions of this article are summarized as follows:

(1) The dual-scale ESE is proposed to estimate the evolutionary state of populations in the RADE, which utilizes the accumulated successful rate and the population diversity across temporal and spatial scales to diagnose optimization requirements, thereby explicitly addressing the *when* issue of EMaTO.

(2) A requirement-driven adaptive knowledge selection (RAKS) strategy is proposed in the RADE. This strategy combines both the evolutionary state similarity and the population distribution similarity to select the most suitable source task as knowledge, efficiently dealing with the *what* issue of EMaTO.

(3) A requirement-driven adaptive knowledge transfer (RAKT) strategy is proposed in the RADE. The strategy adaptively adjusts the knowledge transfer probability (parameter) and selects appropriate transfer operators according to the optimization requirement, efficiently dealing with the *how* issue of EMaTO.

The effectiveness and efficiency of the proposed RADE algorithm are demonstrated through extensive experiments on the CEC19 and WCC122 benchmarks, showing superior performance against five state-of-the-art EMaTO algorithms.

The remainder of this article is organized as follows. Section II provides an overview of EMTO and EMaTO, introduces the related work, and describes the DE algorithm. Section III describes the proposed RADE in detail. Section IV presents the experimental results and discussions. Section V draws the conclusions.

II. PRELIMINARIES

A. MaTOP

A multi-task optimization problem comprises multiple tasks that may be relative or unrelative. Each task can be treated as an independent optimization problem. The goal of solving a multi-task optimization problem with K tasks is to identify a set of K solutions that satisfy:

$$\text{minimize } f(x) = (f_1(x_1), f_2(x_2), \dots, f_K(x_K)) \quad (1)$$

where $f_i(x_i)$ denotes the objective function of the i^{th} task, and x_i is the corresponding decision variable. When $K > 3$, the problem is referred to as a MaTOP.

In MaTOP, the search spaces of different tasks are typically distinct. To facilitate inter-task knowledge transfer, the decision variables of different tasks are first mapped into a unified space using:

$$y_i = \frac{x_i - L_i}{U_i - L_i} \quad (2)$$

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

where L_i and U_i denote the lower and upper bounds of the i^{th} task. The variable y_i represents the normalized x_i within the interval $[0,1]$. During fitness evaluation, y_i is decoded back to the original decision space using the following transformation:

$$x_i = y_i \times (U_i - L_i) + L_i \quad (3)$$

This transformation ensures that the optimization process aligns with each task's specific decision space, enabling effective knowledge transfer across tasks.

B. EMTO and EMaTO

Existing EMTO methods can be broadly categorized into two types: multifactorial optimization (MFO) and multipopulation evolution (MPE).

In MFO, different solutions may be assigned different tasks, but still in the same population and evolving together [22][23]. The multifactorial evolutionary algorithm (MFEA), proposed by Gupta *et al.* [24], is the most representative approach. In MFEA, each solution is given a skill factor indicating the task to which it belongs, enabling knowledge transfer between tasks through the crossover operation involving solutions with different skill factors. Building on the success of MFEA, numerous variants have been proposed. For example, Feng *et al.* [25] combine MFEA with other EC algorithms like particle swarm optimization [26][27] and DE [28][29]. Many researchers enhance knowledge transfer efficacy through strategies like online similarity learning [30], subspace distribution alignment [31], and reinforcement learning-based transfer probability regulation [19].

MPE has also demonstrated impressive performance. In MPE, each population is dedicated to a specific task, and knowledge transfer occurs among different populations through inter-task crossover, usually modulated by a transfer probability [32][33]. MPE research focuses on improving inter-population transfer, simulating task mapping via autoencoders [17], leveraging meta-knowledge for heterogeneous data [34], utilizing anomaly detection to mitigate negative transfer [35], enabling block-level transfer [36], and combining domain adaptation with unified search space [37].

Both MFO and MPE are effective in solving EMTO problems, but they emphasize different aspects. MFO promotes co-optimization within a shared population, while MPE emphasizes independence of different tasks. The distinction significantly affects the scalability of the algorithms. As the number of tasks increasing, the structure of MPE allows the populations to optimize independently and control information sharing probability, which inherently reduces negative task interference. Consequently, the MPE is more suitable for EMaTO that involve a large number of tasks.

Inspired by EMTO algorithms and considering the challenges posed by the increasing number of tasks, many EMaTO algorithms have been proposed. Currently, researchers into EMaTO mainly focus on two aspects: *what* to transfer and *how* to transfer.

For *what* to transfer, researchers have developed diverse methods to quantify the similarity between tasks and guide

knowledge selection. Jiang *et al.* [38] introduced a bi-objective knowledge transfer framework, calculating similarity based on population distribution and solution fitness. Hou *et al.* [39] proposed a bi-space similarity measure, which measured task similarity in both decision space and objective space. Han *et al.* [40] also measured two space similarities but using extended Spearman correlation, linking it to evolutionary states. Dai *et al.* [41] employed a graph attention network to model relationships among multi-source tasks, further improving the modeling capability for complex task interrelations.

For *how* to transfer, Wu *et al.* [42] extended the parameter control mechanism of adaptive DE to EMaTO, and designed an inter-task parameter sharing strategy and a transfer quality assessment module. Li *et al.* [43] dynamically evaluated the performance of genetic algorithm and DE on tasks to select the solver for different tasks and adaptively adjusted the allocation ratio for different solvers. Jiang *et al.* [44] proposed a knowledge transfer method preserving structural consistency. This approach maintains the topological consistency of source task knowledge during transfer, reducing the risk of knowledge distortion. Wu *et al.* [45] utilized evolutionary state and knowledge contribution metrics to adjust transfer strategies.

The above literature shows significant progress on the *what* and *how* issues of knowledge transfer. However, the *when* issue is often handled implicitly, which is unresponsive to the dynamic requirements of the populations directly. Consequently, a clear research gap remains for a framework that explicitly controls the timing of knowledge transfer based on each task's optimization requirement.

C. DE

DE is a widely used EC algorithm known for its simplicity, fast convergence, and robustness across various optimization problems [46][47]. It operates through four main steps: initialization, mutation, crossover, and selection.

First, an initial population of candidate solutions is generated. Then, for each solution $x_i(g)$, a mutant vector $v_i(g)$ is generated as:

$$v_i(g) = x_{r1}(g) + F \times (x_{r2}(g) - x_{r3}(g)) \quad (4)$$

where $x_{r1}(g)$, $x_{r2}(g)$, and $x_{r3}(g)$ are three distinct random solutions from the population, F is the scaling factor, and g denotes the g^{th} generation. This mutation strategy is commonly referred to as 'DE/rand/1'.

The crossover operation in DE is used to combine the mutant $v_i(g)$ with the current solution $x_i(g)$ to create a trial vector $u_i(g+1)$. This process is controlled by the crossover probability CR . For each dimension d , if a random value $\text{rand}(0,1)$ is less than CR or the d matches a randomly pre-chosen dimension q , the trial vector $u_{i,d}(g+1)$ inherits the dimension d from the $v_{i,d}(g)$; otherwise, it inherits from the $x_{i,d}(g)$.

$$u_{i,d}(g+1) = \begin{cases} v_{i,d}(g), & \text{if } \text{rand}(0,1) \leq CR \text{ or } d = q \\ x_{i,d}(g), & \text{otherwise} \end{cases} \quad (5)$$

Selection determines whether the $u_i(g+1)$ will replace the

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

current solution $x_i(g)$ to be the $x_i(g+1)$ in the population. There are two common types of selection strategies, direct selection and environmental selection. For the direct selection, if the fitness value of $u_i(g+1)$ is not worse than the fitness value of $x_i(g)$, the $u_i(g+1)$ will be the $x_i(g+1)$, otherwise, the $x_i(g)$ will be the $x_i(g+1)$. For the environmental selection, all the solutions x_i in the g^{th} generation and all the new generated solutions $u_i(g+1)$ in $(g+1)^{th}$ generation merge together and the best half are selected in to the $(g+1)^{th}$ generation.

III. RADE

A. General Framework of RADE

Fig. 1 depicts the overall framework of RADE, with its pseudo-code described in **Algorithm 1**. To solve a MaTOP involving K tasks, RADE initializes K populations, each dedicated to optimizing a specific task. The RADE algorithm then proceeds to the multi-task process, which comprises three main stages: transfer timing determination, knowledge selection, and knowledge transfer.

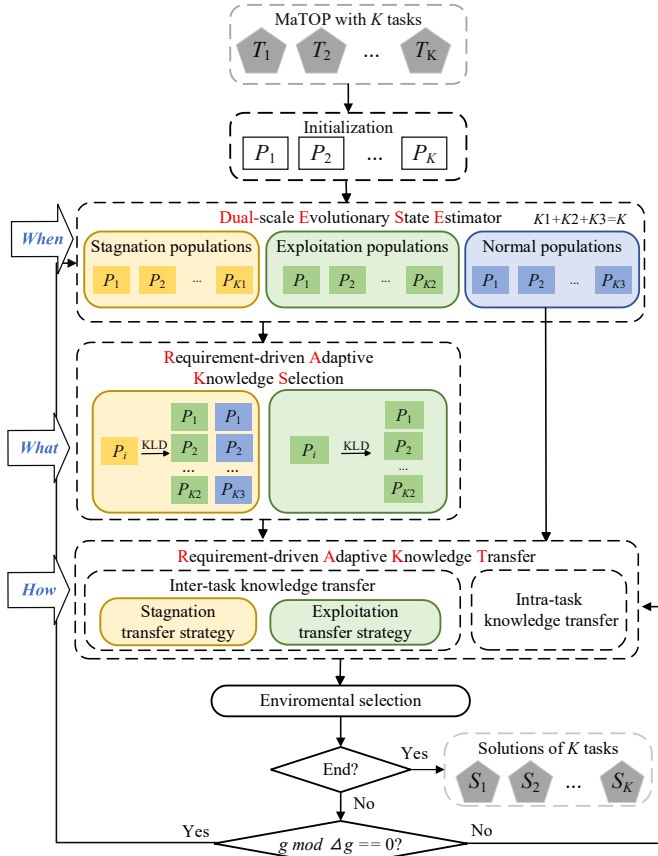


Fig. 1. Flowchart of the RADE framework

In the transfer timing determination stage, the evolutionary state of each population is evaluated from both temporal and spatial scales (**Algorithm 2** in Section III-B), which helps determine whether a population requires knowledge transfer and what kind of knowledge is needed. In the knowledge selection stage, appropriate source tasks are identified based on the evolutionary state similarity and distribution similarity between populations (**Algorithm 3** in Section III-C), ensuring the suitable

knowledge is selected for transfer. Next, in the knowledge transfer stage, the transfer probability and operator are adaptively adjusted (**Algorithm 4** in Section III-D) to facilitate efficient knowledge transfer. Finally, environmental selection retains the optimal solutions from the current population and the new generated population into the next generation.

Algorithm 1: RADE Framework

Input: $T_1, T_2, \dots, T_K$ - the MaTOP with K tasks
Output: $S_1, S_2, \dots, S_K$ - the optimal solutions of task $T_1, T_2, \dots, T_K$
Begin
 1. Initialization;
 2. $g = 1$;
 3. **While** not termination **Do**
 4. **If** $g \bmod \Delta g == 0$ **Then**
 5. Obtain population evolutionary state; // **Algorithm 2** in Sec. III-B
 6. Select the source task by RAKS; // **Algorithm 3** in Sec. III-C
 7. **End If**
 8. **For** $i = 1:K$ **Do**
 9. Update the population by RAKT; // **Algorithm 4** in Sec. III-D
 10. Environmental selection;
 11. **End For**
 12. $g = g + 1$;
 13. **End While**
End

B. Dual-ESE Strategy

We propose a dual-scale method to assess the evolutionary state of a population, thereby determining whether knowledge transfer is needed and what kind of knowledge is required. Fig. 2 illustrates the temporal and spatial information of a population during the evolutionary process. The left part of Fig. 2 is a three-dimensional graph that illustrate the process of a population in solving a two-dimensional single-objective optimization problem. The g represents the evolutionary generation, while the x and y represent the individual position. The blue plane represents the population in a certain generation and the grey plane represents the offspring population before environmental selection. The upper part of the right side of Fig. 2 illustrates the temporal information during the evolution time, showing how the population successfully generates high-quality solutions. For example, the five individuals in the $(g-\Delta g)^{th}$ generation are sorted from up to down (i.e., worse to better) and the five offspring individuals of the $(g-\Delta g)^{th}$ generation are also sorted like this. Then the five better individuals (i.e., two from the current population and three from the offspring population) survive into the $(g-\Delta g+1)^{th}$ generation. The lower part of the right side of Fig. 2 illustrates the spatial information in the search space, showing the distribution and diversity of the population.

Through Fig. 2, we can visually observe the population changes in both the temporal and spatial scales during the optimization process. Since the evolutionary process of the population contains information at both temporal and spatial scales, we design the dual-scale ESE strategy to assess whether knowledge transfer is needed. Specifically, we use the proportion of high-quality solutions accumulated over a period of the evolutionary process and the diversity of the population distribution at a specific generation to achieve the dual-scale assessment of the evolutionary state in both time and space.

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

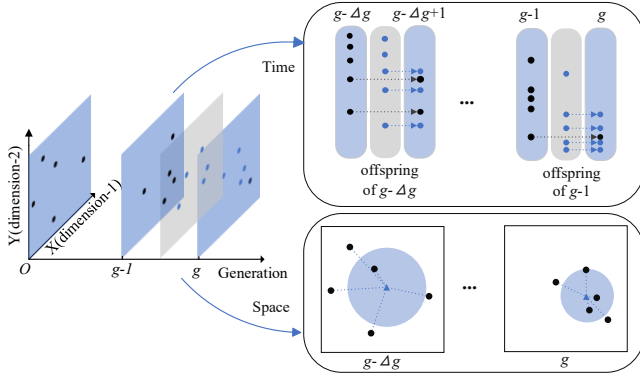


Fig. 2. The temporal and spatial information of a population during the evolutionary process.

However, if we assess the evolutionary state of the population in every generation, it will potentially lead to too much disruption. Therefore, it is preferable to assess the evolutionary state of the population over a period of generations. In order to analyze the optimization requirements of the population based on its evolutionary state, it is necessary to set a detection time window to realize the dynamic state estimation of the population. In this article, we set the time window Δg to 10 generations. When perform the ESE in the every Δg generations, two indicators that corresponding to the temporal and spatial scales are calculated.

First, we use the accumulated success rate to assess the optimization ability of the population in the recent time window. The accumulated successful rate is calculated as:

$$r(g) = \sum_{g-\Delta g \leq g_r < g} \frac{N(g_r)}{NP} \quad (6)$$

where $N(g_r)$ represents the high-quality new solutions produced in the g_r^{th} generation, which have passed environmental selection and progressed to the next generation. This metric reflects the proportion of high-quality solutions among all new solutions produced within the time window Δg . A higher accumulated success rate indicates that the population has a positive tendency to optimize and is more likely to find the optimal solution.

Second, the diversity of the population is used to measure its exploration ability in the search space. The population diversity is calculated as:

$$\Gamma(g) = \frac{1}{NP} \sum_{i=1}^{NP} dist(x_i(g), \mu(g)) \quad (7)$$

where $\mu(g)$ is the mean of the coordinates of solutions in the g^{th} generation and NP is the population size. This metric estimates the distribution or aggregation of populations by calculating the distance between solutions at a given generation. Higher diversity indicates that the population has a greater ability to explore for large search space at the given generation.

By combining these two indicators, we implement the dual-scale ESE assessment of time and space. To better guide the assessment of the population's evolutionary state, two parameters, Θ_r and Θ_d are introduced to normalize the indicators. This dual-scale ESE can be concluded as:

$$\begin{cases} ES = STAGNATION, & \text{if } r(g) \leq \Theta_r, \\ ES = EXPLOITATION, & \text{if } r(g) > \Theta_r, \text{ and } \Gamma(g) \leq \Gamma(g - \Delta g) \times \Theta_d \\ ES = NORMAL, & \text{if } r(g) > \Theta_r, \text{ and } \Gamma(g) > \Gamma(g - \Delta g) \times \Theta_d \end{cases} \quad (8)$$

Accordingly, three possible states need to be analyzed as follows:

(1) **Stagnation state:** When the accumulated successful rate is below the threshold for stagnation detection Θ_r , it indicates that the population has stopped evolving toward the global optimum and is no longer actively searching for the best solution. In this case, the evolutionary state of the population is regarded to be in stagnation, meaning that the population requires a strong disturbance to escape local optima. Suitable knowledge transfer can help adjust the population structure and enhance exploration ability.

(2) **Exploitation state:** When the population shows sufficient improvement of solutions but experiences a notable decrease in spatial diversity compared to the previous detection time, its evolutionary state is regarded as exploitation. This indicates that the population continues to maintain optimization through local search, but faces the risk of a decrease in diversity, which may lead to stagnation. Suitable knowledge transfer can help balance diversity and convergence.

(3) **Normal state:** When the population has good diversity and can produce enough high-quality offspring solutions, it is regarded to have sufficient evolutionary potential to optimize independently. This kind of evolutionary state of the population is classified as normal, which does not need to introduce additional knowledge transfer.

Through the temporal and spatial indicators, the evolutionary state of the population is assessed accurately. Based on this assessment, the optimization requirements of the population are inferred, which can determine whether the population is in a suitable transfer time, and further guide the selection of knowledge and choice of transfer strategy between tasks. This approach ensures that knowledge transfer aligns with the current requirements of each task, thereby improving the efficiency and effectiveness of optimization. **Algorithm 2** provides the pseudo-code of the dual-scale ESE.

Algorithm 2: Dual-scale ESE

Input: K - the number of tasks

$P_1, P_2, \dots, P_K$ - the populations of task $T_1, T_2, \dots, T_K$

Θ_r - threshold for stagnation detection

Θ_d - threshold for diversity assessment

Output: $ES_1, ES_2, \dots, ES_K$ - the evolutionary state of population $P_1, P_2, \dots, P_K$

Begin

1. **For** $i = 1:K$ **Do**

2. Calculate $r(g)$ of P_i according to (6);

3. Calculate $\Gamma(g)$ of P_i according to (7);

4. **If** $r(g) \leq \Theta_r$ **Then**

5. $ES_i \leftarrow STAGNATION$;

6. **Else If** $\Gamma(g) \leq \Gamma(g - \Delta g) \times \Theta_d$ **Then**

7. $ES_i \leftarrow EXPLOITATION$;

8. **Else**

9. $ES_i \leftarrow NORMAL$;

10. **End If**

11. **End For**

End

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

C. RAKS Strategy

To achieve efficient knowledge selection, it is essential to accurately identify the appropriate source tasks for knowledge transfer. The proposed RAKS strategy combine the evolutionary state similarity with the population distribution similarity to select suitable source tasks based on the optimization requirements of populations.

The evolutionary state similarity means that the source population for providing the knowledge is selected among those populations that are with special evolutionary states (may be similar or not similar), according to the requirement of the current population.

For populations in the stagnation state, the optimization requirement is to escape local optima by introducing populations with higher diversity. Therefore, their knowledge selection range is set to include all populations except those are also in stagnation state. These non-stagnant populations exhibit strong performance in generating new solutions, making them suitable as sampling sources for the current population. Thus, these populations are regarded as the potential source tasks to help the current population escape local optima.

For populations in the exploitation state, the focus is on in-depth optimization within localized regions and refining high-quality solutions. Therefore, the source task selection range is set to populations in the same evolutionary state (i.e., also in exploitation state). This ensures the knowledge comes from comparable high-quality domains, enhancing the convergence precision of the population while minimizing the risk of unnecessary interference from excessive information transfer.

After determining the source task selection range according to their evolutionary state similarity, we further use KLD to measure the population distribution similarity between specific tasks and select the most suitable source task for each population. KLD measures the divergence between two probability distributions, and its value can indicate the similarity between these distributions. The KLD between distribution P and distribution Q is defined as

$$KLD(P \parallel Q) = \sum_{x \in X} P(x) \log \frac{P(x)}{Q(x)} \quad (9)$$

where P represents the distribution of the current population and Q represents the distribution of the candidate source population, with X being the sample place. A smaller KLD value indicates higher similarity between the two distributions, suggesting that their search behavior in the solution space might be more consistent. By calculating the KLD between the current population and the potential source populations, we can identify which task shares the most similar distributions with the current population, thereby informing the selection of source tasks for knowledge transfer.

Due to the potential optimal solutions for different tasks may locate in different regions of the unified search space, a direct KLD calculation could be misleading. The KLD metric is sensitive to differences in both mean (position) and covariance (shape). A large distance between the mean positions of each population would dominate the KLD value, masking any underlying similarity in the covariance structures of populations.

To improve the accuracy of the KLD for RAKS, we first align the populations by shifting the source population to the center of the target (i.e., current) population. The normalized coordinate $\tilde{x}_i$ is calculated as:

$$\tilde{x}_i = x_i - \mu_s + \mu_t \quad (10)$$

where μ_s and μ_t are the mean coordinates of the source and target populations, respectively. This transformation brings the source knowledge into a relevant region for the target task, reducing biases caused by positional differences and thereby improving the accuracy of the KLD calculation [34][38]. After normalization, we calculate the KLD to ensure a more precise assessment as:

$$KLD(P_i \parallel \tilde{P}_s) = tr(\Sigma_s^{-1} \Sigma_t) + (\mu_s - \tilde{\mu}_t)^T \Sigma_s^{-1} (\mu_t - \tilde{\mu}_s) - D + \log\left(\frac{\det(\Sigma_s)}{\det(\Sigma_t)}\right) \quad (11)$$

where Σ_t and Σ_s are the covariance matrices of P_i and $\tilde{P}_s$, respectively, D represents the dimension of the covariance matrices. $tr(\cdot)$ and $\det(\cdot)$ are the trace and determinant of the computational matrix. It is noted that this formula assumes the populations follow a Gaussian distribution. This is a common and practical assumption in the EC community that allows for a computationally efficient estimation of distributional similarity based on the mean and covariance of the positions of the solutions in the population [11][48].

Algorithm 3 shows the pseudo-code of the RAKS strategy. According to the optimization requirement, this strategy sets a suitable range of source tasks for each population to select the knowledge, considering both the evolutionary state similarity and the population distribution similarity.

Algorithm 3: RAKS

Input: $P_1, P_2, \dots, P_K$ – the population of task $T_1, T_2, \dots, T_K$
 $ES_1, ES_2, \dots, ES_K$ – the evolutionary state of population $P_1, P_2, \dots, P_K$
Output: $T_{s1}, T_{s2}, \dots, T_{sK}$ – the source task for population $P_1, P_2, \dots, P_K$
Begin
1. **For** $i = 1:K$ **Do**
2. **If** $ES_i = STAGNATION$ **Then**
3. Calculate KLD of P_i with *NORMAL* and *EXPLOITATION*;
4. **Else If** $ES_i = EXPLOITATION$ **Then**
5. Calculate KLD of P_i with *EXPLOITATION*;
6. **End If**
7. Select the best task with minimal KLD of P_i to be T_{si} ;
8. **End For**
End

D. RAKT Strategy

To address the specific requirements of different populations during the evolutionary process, we propose the adaptive RAKT strategy. The main idea is to provide optimal evolutionary support through adaptive transfer parameters and adaptive transfer operators, thereby enhancing knowledge adaptability and transfer efficiency. The pseudo-code of the RAKT strategy is presented in **Algorithm 4**.

(1) **Adaptive transfer parameter:** Intuitively, populations in different evolutionary states have different requirements for inter-task knowledge transfer probability (i.e., the transfer parameter).

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

For the population in the stagnation state, which urgently needs diversity, a higher inter-task transfer probability is necessary to help it escape local optima. For the population in the exploitation state, lowering inter-task transfer probability reduces external knowledge interference, allowing the population to focus on refining high-quality solutions.

Specifically, we regulate the inter-task transfer probability using the parameter rpm and introduce a balancing factor, denoted as α , to adaptively change the inter-task transfer probability more suitable for different optimization requirements. The baseline value of 0.7 is empirically determined, following the setting in [8]. The adjustment mechanism is formalized as:

$$\begin{cases} rpm = 0.7 + \alpha, & \text{if } ES = STAGNATIONS \\ rpm = 0.7 - \alpha, & \text{if } ES = EXPLOIATION \end{cases} \quad (12)$$

(2) Adaptive transfer operator: To further enhance the effectiveness of knowledge transfer, populations with different optimization requirements not only vary the parameter rpm but also employ distinct transfer operators. The RAKT strategy includes intra-task transfer and inter-task transfer.

Intra-task transfer: The population in the normal state is regarded capable of evolving independently, and thus, it does not need to introduce knowledge of other tasks to help evolution. For such populations, intra-task knowledge transfer between solutions is emphasized. Therefore, we use DE/rand/1 operator via (4) to help the population fully explore the search space. The details of the knowledge transfer for population in normal state are illustrated in Lines 2-7 in **Algorithm 4**.

Inter-task transfer: If the population in stagnation state or exploitation state, it means the population is with insufficient activity to a varying degree. Therefore, it not only needs to stimulate the activity from intra-task knowledge transfer but also needs inter-task knowledge transfer to help the population improve the activity.

To efficiently execute the inter-task knowledge transfer, we introduce the shift mechanism for the source task. Specifically, after a source task T_s has been selected according to the RAKS strategy, its corresponding population P_s is shifted towards the center of the target population P_i , denoted as $\tilde{P}_s$ like in (10). This alignment is a crucial step for effective knowledge transfer. As a large positional gap between the source and target populations may hinder the mutation operator; transferring a solution from a distant source may generate offspring far from the promising search regions of the target population, leading to inefficient transfer. The shifted population $\tilde{P}_s$ can ensure that the transferred knowledge is introduced within a relevant context, significantly improving the probability of generating high-quality offspring [34][38].

For the population in the stagnation state, the inter-task transfer operator is based on the Cauchy distribution [49] to generate new solutions. The long tail characteristics of the Cauchy distribution can produce significant perturbations, helping to overcome the limitations imposed by stagnation and introducing new variations to increase population diversity. Specifically, the mutation operator generates the mutation vector $v_i(t)$ from the Cauchy distribution as:

$$v_i(g) = \text{cauchy}(\mu_a, 0.1) \quad (13)$$

where μ_a is the mean value of the union set $\tilde{P}_s \cup P_i$ defined as:

$$\mu_a = \frac{1}{|\tilde{P}_s \cup P_i|} \sum_{x_a \in \tilde{P}_s \cup P_i} x_a \quad (14)$$

This operator generates new solutions based on the current solutions with strong perturbations, helping to escape local optima and explore new search directions. However, this operator for inter-task transfer is controlled by the parameter rpm , as Line 10 in **Algorithm 4**. If intra-task knowledge transfer is performed for the stagnant population, it generates offspring individual like the normal population, i.e., use DE/rand/1 operator via (4). The details of the knowledge transfer for population in stagnation state are illustrated in Lines 8-19 in **Algorithm 4**.

Algorithm 4: RAKT

Input: P_i the current population of task T_i

P_s the selected source population of task T_i

Output: O_i the offspring of population P_i

Begin

```

1.  $Q = \emptyset$ ;
2. If  $ES_i == NORMAL$  Then
3.   For  $j = 1:NP$  Do // only intra-task knowledge transfer
4.     Randomly select three parents  $x_{r1}$ ,  $x_{r2}$ , and  $x_{r3}$  from  $P_i$ ;
5.     Generate  $OP_j$  via 'DE/rand/1' in (4);
6.     Add the offspring  $OP_j$  into  $O_i$ ;
7.   End For
8. Else If  $ES_i == STAGNATION$  Then
9.   For  $j = 1:NP$  Do
10.    If  $\text{rand}(0,1) < rpm$  Then
11.      Shift  $P_s$  to the center of  $P_i$  via (10) as  $\tilde{P}_s$ ;
12.      Calculate mean of  $\tilde{P}_s \cup P_i$  via (14);
13.      Generate  $OP_j$  via Cauchy distribution in (13);
14.    Else
15.      Randomly select three parents  $x_{r1}$ ,  $x_{r2}$ , and  $x_{r3}$  from  $P_i$ ;
16.      Generate  $OP_j$  via 'DE/rand/1' in (4);
17.    End If
18.    Add the offspring  $OP_j$  into  $O_i$ ;
19.  End for
20. Else If  $ES_i == EXPLOITATION$  Then
21.   For  $j = 1:NP$  Do
22.    If  $\text{rand}(0,1) < rpm$  Then
23.      Shift  $P_s$  to the center of  $P_i$  via (10) as  $\tilde{P}_s$ ;
24.      Calculate mean of  $\tilde{P}_s \cup P_i$  via (13);
25.      Generate  $OP_j$  via 'DE/current-to-best/1' in (15);
26.    Else
27.      Randomly select three parents  $x_{r1}$ ,  $x_{r2}$ , and  $x_{r3}$  from  $P_i$ ;
28.      Generate  $OP_j$  via 'DE/rand/1' in (4);
29.    End If
30.    Add the offspring  $OP_j$  into  $O_i$ ;
31.  End For
32. End If
End

```

For the population in the exploitation state, the inter-task transfer operator is based on 'DE/current-to-best/1' mutation strategy, leveraging the information of the best solution in the current population. The mutation operation is as:

$$v_i(g) = x_i(g) + F \times (x_{best}(g) - x_i(g)) + F \times (x_{r1}(g) - x_{r2}(g)) \quad (15)$$

where x_{best} is the best solution in the current population, x_i is the current solution, and x_{r1} and x_{r2} are randomly selected solutions

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

from $\tilde{P}_s \cup P_i$. The ‘DE/current-to-best/1’ strategy uses the best solution as the basis for mutation to maintain optimization stability and introduces variations based on differences between solutions, further refining the high-quality solutions found. This operator ensures precision in optimization with the current solution space and effectively reduces interference from excessive information transfer.

However, the same as the stagnant population, this operator for inter-task transfer is also controlled by the parameter rm_p , as Line 21 in **Algorithm 4**. If intra-task knowledge transfer is performed for the exploitation population, it generates offspring individual like the normal population, i.e., use DE/rand/1 operator via (4). The details of the knowledge transfer for population in exploitation state are illustrated in Lines 20-32 in **Algorithm 4**.

E. Computational Complexity Analysis

To provide a theoretical analysis of the computational time and space complexity of the proposed RADE algorithm, we define the following variables: K is the number of tasks, NP is the population size per task, D is the dimension of the problem, and the time window Δg is the generation interval for state estimation.

(1) **Time complexity:** The overall time complexity is analyzed by considering the baseline DE framework and the additional overhead introduced by core components of RADE. Support that the standard multi-population DE algorithm deals with K tasks, the algorithm has a computational complexity of $O(K \times NP \times D)$ per generation. The additional overhead in RADE stems from the dual-scale ESE, RAKS, and RAKT.

Dual-scale ESE: This module is activated every Δg generations. Its main cost comes from calculating the population diversity for all K tasks, which has a complexity of $O(K \times NP \times D)$. Thus, its complexity per generation is $O(K \times NP \times D / \Delta g)$.

RAKS strategy: This module is also activated every Δg generations and represents the most significant computational overhead. For a task requiring knowledge transfer, it calculates the KLD against a set of K_s candidate source tasks. A single KLD calculation is dominated by computing covariance matrices $O(NP \times D^2)$ and subsequent matrix operations $O(D^3)$. In the worst-case scenario where K_t tasks require transfer and each compares against K_s candidates, the complexity per activation is $O(K_t \times K_s \times (NP \times D^2 + D^3))$. The complexity per generation is therefore $O(K_t \times K_s \times (NP \times D^2 + D^3) / \Delta g)$.

RAKT strategy: The main overhead in each generation is the alignment of source populations for tasks performing inter-task transfer. This has a complexity of $O(K_t \times NP \times D)$ per generation, as K_t tasks require transfer.

In summary, the total time complexity per generation is the sum of the baseline and the additional overhead of these components. As the RAKS strategy is the dominant factor, the total time complexity of RADE can be considered similar to that of RAKS, particularly for problems with high dimensionality or a large number of tasks. However, it is crucial to note that this peak cost occurs only periodically and often for a subset of tasks ($K_t < K$), rather than being a constant overhead in every generation.

(2) **Space complexity:** The primary space requirement of RADE is storing the K populations, each of size NP with solutions of dimension D . This results in a baseline space complexity of $O(K \times NP \times D)$. RADE introduces minor additional and temporary space requirements. The RAKS strategy stores covariance matrices of size $D \times D$ (i.e., $O(D^2)$), and the RAKT strategy creates a temporary copy of a source population ($O(NP \times D)$). Since these additional storages are temporary and of a lower order than the baseline requirement for storing all populations, the overall space complexity of RADE remains $O(K \times NP \times D)$.

IV. EXPERIMENTAL STUDIES

In this section, we present the experimental evaluation of RADE. First, we assess the effectiveness of the proposed RADE by comparing it with the advanced EMaTO algorithm using the CEC19 MaTOPs test suite. Next, we test RADE performance on the more challenging WCCI22 MaTOPs test suite. We then analyze how the main components affect the performance of the RADE. Finally, we examine how different parameter settings affect the performance of RADE.

A. Benchmark Functions and Performance Metrics

The experiments are conducted on the CEC19 competition on EMaTO [50] (denoted as CEC19) and the WCCI22 competition on EMaTO’s single-target MaTOP [51] (denoted as WCCI22). The CEC19 benchmark consists of six MaTOPs, each with 50 homogeneous generated by translating and rotating single-target base functions. The WCCI22 benchmark consists of 10 MaTOPs, each of which also includes 50 tasks. There are both isomorphic and homogeneous tasks in WCCI22, making it a more complex and challenging test suite. The basic functions used in CEC19 and WCCI22 are summarized in Table S.I in the supplemental material.

In this study, two indicators are used to evaluate the performance of the algorithms. The first indicator is the average fitness value (AFV). The AFV represents the mean of the best fitness values obtained for the K tasks. Given the fitness values $f_1, f_2, \dots, f_K$ for each task, AFV is calculated as:

$$AFV(f_1, f_2, \dots, f_K) = \frac{1}{K} \sum_{i=1}^K f_i \quad (16)$$

Since each task in the CEC19 and WCCI22 test suites is a minimization problem, a smaller AFV indicates better performance for the corresponding algorithm.

The second indicator is the mean standard score (MSS) [52][53]. Given that the fitness values of different tasks may vary significantly in scale, the AFV could be disproportionately influenced by tasks with larger fitness values. To avoid the impact of differing fitness scales, MSS standardizes the fitness values before averaging them, which is calculated as:

$$MSS(f_1, f_2, \dots, f_K) = \frac{1}{K} \sum_{i=1}^K \frac{f_i - \bar{f}_i}{\delta_i} \quad (17)$$

where $\bar{f}_i$ and δ_i are the mean and standard deviation of the fitness value for task T_i . By standardizing the fitness values, MSS provides a more balanced and comprehensive assessment of

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

algorithm performance, ensuring that tasks with smaller fitness values are not overshadowed by those with larger values. Also, similar to the AFV, a smaller MSS indicates that the algorithm has better performance.

To statistically analyze the results, we employ Wilcoxon's rank-sum test [54] at a significance level of $\alpha = 0.05$. The notations “+/-” denote that results obtained by RADE are “significantly superior/equivalent/inferior” to those obtained by the compared algorithms. To further confirm the performance of RADE, a Friedman test [55] with $\alpha = 0.05$ is also performed, and the average ranks of all algorithms across all benchmark problems are compared.

B. Compared Algorithms and Parameter Setting

Five state-of-the-art algorithms, which are MaTDE [48], ASCMFDE [31], MTEA_SaO [43], BoKT-DE [38], and TRADE [42], are adopted as the comparison algorithms.

The parameter settings for RADE are listed in Table I. For MaTDE, ASCMFDE, MTEA_SaO, BoKT-DE, and TRADE, the population size is set to 100. The population size of ASCMFDE is 5000 since ASCMFDE utilizes a single population framework. The remaining parameter settings follow the configurations specified in their original papers. Each algorithm is executed 20 times independently for obtaining the average experimental results.

C. Results and Discussions

The AFV and MSS obtained by RADE, MaTDE, ASCMFDE, MTEA-SaO, BoKT-DE, and TRADE on CEC19 and WCCI22 test suites are presented in Table II. The results of the best-performing algorithms on each problem are shown in **boldface**.

As can be seen in Table II, for AFV, the proposed RADE obtains the best results on one CEC19-MaTOP and eight WCCI22-MaTOPs, including CEC19-P5, and WCCI22-P1, WCCI22-P3, WCCI22-P5~WCCI-P10. The statistical results by Wilcoxon's rank-sum show that the optimization performance of RADE is superior to other compared algorithms in terms of both AFV and MSS. Moreover, the Friedman test results on the AFV values are shown in Table S.II in the supplementary material. The

TABLE I
PARAMETER SETTINGS FOR RADE

Parameter	Value
maximum number of evaluations $maxFEs$	5000000
scaling factor F	0.7
crossover rate CR	0.5
threshold for stagnation detection Θ_r	0.1
threshold for diversity assessment Θ_d	0.9
knowledge transfer balancing coefficient α	0.2
time window Δg	10

mean rank of RADE is 2.3750, which is the best overall rank, indicating its superior performance across the benchmarks.

Through analysis of these results, we find RADE's main strength lies in solving complex problems that contain diverse tasks, achieving the best results on six out of a total of seven heterogeneous problems (i.e., the WCCI22-P4~WCCI22-P10). That's because the dual-scale ESE can accurately detect the unique optimization state of each population (i.e., task) and determine whether it is in the right transfer time. This precise requirement-driven transfer timing detection makes RADE stand out from traditional EMaTO algorithms in heterogeneous problems. The RAKS and RAKT strategies can dynamically match the requirements of different tasks, further strengthening its cross-task adaptability, making RADE a powerful algorithm for solving complex heterogeneous MaTOPs.

In order to show the performance of each algorithm more directly and analyze the optimization process, some representative convergence curves on homogeneous and heterogeneous problems are given in Fig. 3, and complete convergence curves for all benchmark problems are in Fig. S.1 and Fig. S.2 in the supplemental material. According to Fig. 3(a), (c), (d), RADE can maintain a stable convergence trend and continuously improve the quality of the solution while the convergence speed of the comparison algorithms slows down significantly. This phenomenon strongly demonstrates the effectiveness of the ‘requirement-driven’ framework. Its dual-scale ESE strategy can accurately identify when the population is weak in evolution by itself, and introduce suitable knowledge in

TABLE II
EXPERIMENTAL RESULTS OF AFV AND MSS OBTAINED BY RADE, MATDE, ASCMFDE, MTEA_SAO, BoKT-DE, AND TRADE ON CEC19-MATOPs AND WCCI22-MATOPs

Problem	RADE		MaTDE		ASCMFDE		MTEA SaO		BoKT-DE		TRADE	
	AFV	MSS	AFV	MSS	AFV	MSS	AFV	MSS	AFV	MSS	AFV	MSS
CEC19-P1	9.61E+01	-1.55E-01	1.97E+02+	5.46E-01+	7.15E+02+	-3.93E-01-	2.52E+02+	6.63E-01+	8.86E+01+	-1.95E-01=	4.20E+01-	-4.67E-01-
CEC19-P2	7.41E-01	-6.44E-01	2.95E-05-	-7.22E-01-	2.02E+01+	1.40E+00+	2.01E-03-	-7.22E-01-	2.02E+01+	1.40E+00+	5.04E-08-	-7.22E-01-
CEC19-P3	2.54E+02	-2.87E-01	3.20E+02+	2.81E-01+	4.04E+02+	1.01E+00+	1.09E+02-	-1.57E+00-	3.40E+02+	4.54E-01+	3.01E+02+	1.10E-01+
CEC19-P4	3.43E-04	-3.67E-01	7.09E-03+	1.20E+00+	2.07E-03+	7.72E-02+	1.72E-03+	-1.76E-02+	1.51E-05-	-4.50E-01-	5.47E-05-	-4.40E-01-
CEC19-P5	2.97E-03	-5.85E-01	2.35E-01+	7.05E-01+	3.04E-01+	1.13E+00+	5.75E-03+	-5.66E-01+	1.81E-02+	-4.98E-01+	7.23E-02+	-1.87E-01+
CEC19-P6	8.92E+03	4.66E-01	2.88E-03-	-1.45E+00-	1.10E+04+	8.45E-01+	1.43E+03-	-1.15E+00-	1.17E+04+	9.98E-01+	8.31E+03-	2.91E-01=
WCCI22-P1	1.80E-25	-4.38E-01	1.28E-06+	-3.79E-01+	5.33E-05+	2.03E+00+	1.07E-05+	-3.39E-01+	2.07E-08+	-4.37E-01+	1.02E-13+	-4.38E-01+
WCCI22-P2	4.53E+02	3.29E-01	1.39E+02-	-1.71E-01-	7.85E+01-	-2.66E-01-	5.95E+02=	5.06E-01=	1.30E+02-	-1.81E-01-	1.09E+02-	-2.17E-01-
WCCI22-P3	2.46E+02	-2.53E-01	2.70E+02+	-5.28E-02+	4.04E+02+	1.12E+00+	1.08E+02-	-1.48E+00-	3.40E+02+	5.63E-01+	2.88E+02+	1.04E-01+
WCCI22-P4	2.16E+02	-2.99E-01	1.27E+02-	5.12E-01+	1.19E+02-	5.34E-01+	2.74E+02+	-1.44E-01+	4.51E+02+	-2.00E-01+	5.34E+01-	-4.03E-01-
WCCI22-P5	2.40E+01	-8.16E-01	1.21E+02+	8.93E-01+	1.35E+02+	4.34E-01+	4.42E+01+	7.02E-02+	9.43E+01+	-3.16E-01+	1.01E+02+	-2.66E-01+
WCCI22-P6	2.17E+03	-6.46E-01	4.65E+03+	6.16E-01+	4.29E+03+	4.36E-01+	2.54E+03+	-3.13E-01+	4.52E+03+	3.56E-02+	4.16E+03+	-1.28E-01+
WCCI22-P7	2.48E+01	-7.86E-01	1.22E+02+	1.28E+00+	1.37E+02+	9.20E-03+	4.49E+01+	7.86E-02+	8.92E+01+	-3.32E-01+	9.67E+01+	-2.47E-01+
WCCI22-P8	1.20E+02	-5.86E-01	3.58E+02+	1.13E+00+	1.59E+02+	9.22E-02+	1.51E+02=	-5.30E-02+	1.04E+03+	-2.81E-01+	9.75E+01=	-2.97E-01+
WCCI22-P9	8.60E+02	-7.08E-01	2.23E+03+	1.01E+00+	2.07E+03+	2.48E-01+	1.02E+03+	-2.56E-01+	2.57E+03+	-1.20E-01+	1.94E+03+	-1.79E-01+
WCCI22-P10	1.00E+03	-8.64E-01	2.67E+03+	1.19E+00+	2.49E+03+	3.67E-01+	1.14E+03+	-3.25E-01+	2.53E+03+	-2.20E-01+	2.48E+03+	-1.51E-01+
+/-=	NA	NA	12/4/0	13/3/0	14/2/0	14/2/0	10/4/2	11/4/1	13/2/1	13/2/1	9/5/2	10/5/1

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

time through the RAKT strategy to help it by transferring knowledge from other tasks. In contrast, Fig. 3(b) shows that RADE performs worse than MaTDE on CEC19-P6. The main reason may be due to that the PCA-based subspace alignment strategy of MaTDE is particularly suitable to solve the problem generated by the rotation of Schwefel functions.

The runtime analysis provides deeper insight into RADE's adaptive nature. The runtime results are presented in Table S.III in the supplementary material. According to Table S.III, the values of the mean rank of RADE is 2.1250, which is only slightly beaten by TRADE. As the overall runtime rank of RADE is highly competitive, its computational efficiency demonstrates stronger competitiveness on heterogeneous problems than on homogeneous ones. This is a direct consequence of its adaptive mechanism. On homogeneous problems, a large number of tasks tend to enter the same evolutionary state simultaneously. This expands the candidate pool for knowledge sharing within the RAKS strategy, triggering more of the computationally intensive KLD calculations and thus increasing its runtime relative to simpler algorithms. Conversely, on heterogeneous problems where task states are more diverse, the specific computational overhead of KLD is naturally reduced, allowing the efficiency of RADE to be more competitive.

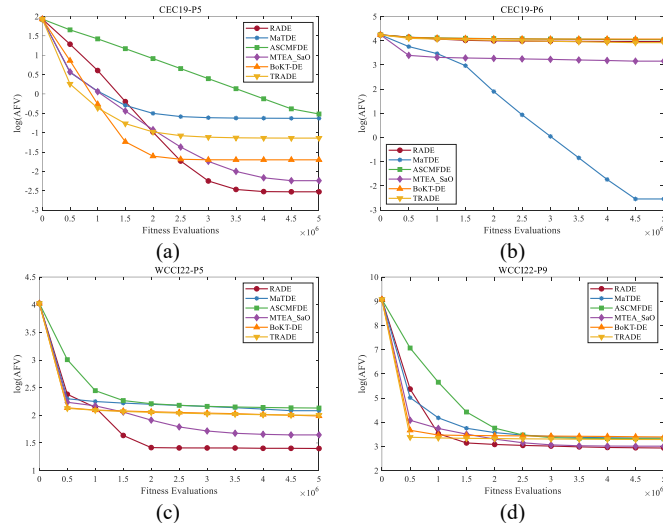


Fig. 3. Convergence curves of $\log(\text{AFV})$ obtained by RADE, MaTDE, ASCMFDE, MTEA-SaO, BoKT-DE, and TRADE on CEC19-P5, CEC19-P6, WCCI22-P5, and WCCI22-P9.

This behavior demonstrates that the computational cost of RADE is not static but dynamically adapts to the problem structure. It strategically allocates more computational resources when tasks are similar and frequent transfer is potentially beneficial, while conserving resources when tasks are diverse and indiscriminate transfer is risky. This makes its computational cost a justifiable and intelligent trade-off for its superior effectiveness in complex problems.

In general, RADE performs well in both efficiency and effectiveness, in terms of the optimization results of AFV and MSS, the convergence curves, and the runtime.

D. Ablation Study of Component Effects

In this section, we analyze the effectiveness of the requirement-

driven strategy by analyzing the three main components of RADE, dual-scale ESE, RAKS, and RAKT, and discuss their specific effectiveness.

First, we analyze the dual-scale ESE strategy, which involves an optimization requirement for knowledge selection and transfer. As the dual-scale ESE strategy is combined with the time scale and space scale, we introduce two variants of RADE: RADE without time scale estimation and RADE without space scale estimation. The experimental results of the AFV obtained by RADE and its two variants are presented in Table III. The results show that RADE obtains the best results on 11 out of the 16 problems, indicating that both the time scale and space scale are important to the estimation of evolutionary state and the detection of optimization requirement.

TABLE III
EXPERIMENTAL RESULTS OF AFV OBTAINED BY RADE AND RADE WITHOUT TIME SCALE ESTIMATION OR SPACE SCALE ESTIMATION

Problem	RADE	RADE without time scale estimation	RADE without space scale estimation
CEC19-P1	9.61E+01	1.40E+02+	4.37E+02+
CEC19-P2	7.41E-01	2.00E+01+	4.57E+00+
CEC19-P3	2.54E+02	3.46E+02+	2.97E+02+
CEC19-P4	3.43E-04	5.13E-04=	7.69E-03+
CEC19-P5	2.97E-03	5.23E-03=	7.82E-01+
CEC19-P6	8.92E+03	1.07E+04+	8.60E+03-
WCCI22-P1	1.80E-25	2.06E-25+	2.09E-02+
WCCI22-P2	4.53E+02	6.07E+02=	2.23E+03+
WCCI22-P3	2.46E+02	3.46E+02+	2.98E+02+
WCCI22-P4	2.16E+02	1.88E+02=	9.05E+02+
WCCI22-P5	2.40E+01	2.25E+01=	1.03E+02+
WCCI22-P6	2.17E+03	2.35E+03+	3.36E+03+
WCCI22-P7	2.48E+01	2.18E+01-	9.80E+01+
WCCI22-P8	1.20E+02	1.75E+02=	1.48E+02=
WCCI22-P9	8.60E+02	9.10E+02=	9.74E+02+
WCCI22-P10	1.00E+03	9.82E+02=	1.45E+03+
+/-/=	-/-/-	7/1/8	14/1/1

Second, we examine whether the optimization requirement positivity affects source task selection. To explore this, we introduce two variants of RADE: RADE without the selection range and RADE without KLD similarity, and the experimental results of the AFV obtained by RADE and its two variants are presented in Table IV. The results show that RADE obtains the best results on 12 out of the 16 problems. Therefore, setting the selection range based on optimization requirements prior to KLD analysis is both meaningful and efficient. The RAKS strategy not only decreases computational complexity but also maintains the strong performance of the algorithm.

TABLE IV
EXPERIMENTAL RESULTS OF AFV OBTAINED BY RADE AND RADE WITHOUT SELECTION RANGE OR KLD SIMILARITY

Problem	RADE	RADE without selection range	RADE without KLD similarity
CEC19-P1	9.61E+01	1.75E+02+	1.39E+02+
CEC19-P2	7.41E-01	3.10E+00+	1.12E+00+
CEC19-P3	2.54E+02	2.70E+02+	2.58E+02=
CEC19-P4	3.43E-04	1.09E-03+	5.89E-03+
CEC19-P5	2.97E-03	5.85E-02+	6.93E-03+
CEC19-P6	8.92E+03	8.67E+03+	9.18E+03+
WCCI22-P1	1.80E-25	7.54E-06=	2.67E-25=

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

WCCI22-P2	4.53E+02	5.58E+02=	5.20E+02=
WCCI22-P3	2.46E+02	2.77E+02+	2.65E+02+
WCCI22-P4	2.16E+02	2.24E+02=	2.11E+02=
WCCI22-P5	2.40E+01	3.02E+01+	3.41E+01+
WCCI22-P6	2.17E+03	2.50E+03+	2.50E+03+
WCCI22-P7	2.48E+01	2.48E+01=	2.33E+01=
WCCI22-P8	1.20E+02	5.90E+02+	2.50E+02=
WCCI22-P9	8.60E+02	1.08E+03+	8.96E+03=
WCCI22-P10	1.00E+03	1.03E+03=	9.98E+02=
+/-/=	-/-/-	11/0/5	9/0/7

Third, we evaluate the effectiveness of the RAKT in facilitating positive knowledge transfer between tasks. We create two variants of RADE, each using a single, fixed transfer probability and operator. Specifically, these variants do not consider the optimization requirements of the population during transfer, and we call these two variants RADE-Cauchy and RADE-current-to-best. The RADE-Cauchy variant means that no matter what the population state is, the *rmu* is set fixed as $0.7+\alpha$ and the operator is fixed as Cauchy mutation as (13). The RADE-current-to-best variant means that no matter what the population state is, the *rmu* is set fixed as $0.7-\alpha$ and the operator is fixed as 'DE/current-to-best/1' as (15). The experimental results of the AFV obtained by RADE and two variants with a single knowledge transfer strategy are shown in Table IV. According to Table V, RADE obtains the best results on 15 out of the 16 problems, which strongly confirms the efficiency of RAKT.

TABLE V
EXPERIMENTAL RESULTS OF AFV OBTAINED BY RADE AND RADE WITH SINGLE KNOWLEDGE TRANSFER STRATEGY

Problem	RADE	RADE-Cauchy	RADE-current-to-best
CEC19-P1	9.61E+01	2.45E+03+	4.09E+02+
CEC19-P2	7.41E-01	1.63E+00+	2.02E+01+
CEC19-P3	2.54E+02	4.46E+02+	3.38E+02+
CEC19-P4	3.43E-04	2.30E-01+	7.15E-03+
CEC19-P5	2.97E-03	7.96E+00+	7.46E-01+
CEC19-P6	8.92E+03	7.93E+03+	1.07E+04+
WCCI22-P1	1.80E-25	1.02E+00+	3.41E-02+
WCCI22-P2	4.53E+02	1.19E+03+	4.56E+03+
WCCI22-P3	2.46E+02	4.07E+02+	3.19E+02+
WCCI22-P4	2.16E+02	3.63E+03+	1.99E+03+
WCCI22-P5	2.40E+01	1.69E+02+	2.87E+01+
WCCI22-P6	2.17E+03	5.77E+03+	3.04E+03+
WCCI22-P7	2.48E+01	1.71E+02+	2.84E+01+
WCCI22-P8	1.20E+02	2.56E+04+	1.46E+03+
WCCI22-P9	8.60E+02	2.33E+04+	1.50E+03+
WCCI22-P10	1.00E+03	1.13E+03+	1.05E+03=
+/-/=	-/-/-	11/0/5	9/0/7

E. Sensitivity Analysis of the Parameters

The performance of RADE depends on four key parameters: the knowledge transfer balancing coefficient α , the time window Δg , the stagnation detection threshold Θ_r , and the diversity assessment threshold Θ_d . This section presents a sensitivity analysis to evaluate the impact of these parameters on the algorithm's performance and to justify the values used in our experiments. The results of RADE using different α , Δg , Θ_r , and Θ_d are shown in Fig. 4.

The parameter α balances the transfer probability for populations with different requirements. As shown in Table S.IV in the supplemental material, RADE achieves the best overall performance when $\alpha = 0.2$ (mean rank 1.8125). A value that is too small provides insufficient differentiation in transfer intensity, while a value that is too large (e.g., 0.25) can cause excessive disruption. Therefore, α is set to 0.2 in this article.

The time window Δg determines the frequency of the evolutionary state assessment. According to the experimental results presented in Table S.V in the supplemental material, the algorithm performs best at $\Delta g = 10$. In the dual-scale ESE, the time scale needs to be measured by the accumulated success rate, so we consider that $\Delta g = 5$ will make the evaluation interval too short, making the algorithm sensitive to noise. However, when $\Delta g = 50$, it will lead to the delay of requirement detection, miss the best transfer time, and cannot respond to the requirements of the population, and make adaptive transfer strategy adjustments in time. Therefore, we select $\Delta g = 10$.

The thresholds Θ_r and Θ_d are central to the dual-scale ESE. For the stagnation threshold Θ_r , the results are in Table S.VI in the supplemental material, which show that the algorithm performance is strongest at $\Theta_r = 0.1$ (mean rank 1.9375). This aligns with its definition, as only a very low success rate should trigger a stagnation state. For the diversity threshold Θ_d , the results in Table S.VII in the supplemental material indicate that high values are optimal, with $\Theta_d = 0.9$ achieving the best results in 7 problems and $\Theta_d = 0.95$ also achieving the best results in 7 problems. Therefore we select $\Theta_d = 0.9$ as it obtains the best mean rank value 1.6250.

In summary, these analyses demonstrate that RADE's performance is robust across a reasonable range for its key parameters, and the settings used in our comparative study are well-supported by the experiments.

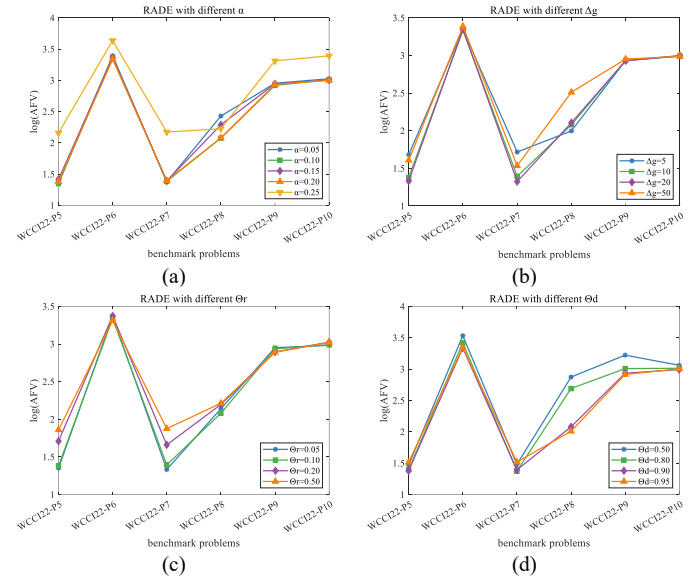


Fig. 4. Result of $\log(\text{AFV})$ obtained by RADE with different parameters on WCCI22-P5, WCCI22-P6, WCCI22-P7, WCCI22-P8, WCCI22-P9 and WCCI22-P10.

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

V. CONCLUSION

In this article, we propose the RADE algorithm for many-task optimization. By introducing the dual-scale ESE that evaluates population performance from both temporal and spatial scales, RADE implements the precise detection of transfer timing and effectively identifies the optimization requirement of each population. Then we further present the RAKS strategy, which integrates evolutionary state similarity and population distribution strategy to select the most suitable knowledge source. Additionally, the RAKT strategy, featuring adaptive knowledge transfer probability and operators, is designed to optimize the transfer process dynamically throughout the evolution process. Experimental results on the CEC19 and WCCI22 benchmarks show that the RADE outperforms existing state-of-the-art EMaTO methods in terms of both effectiveness and efficiency.

Through this work, we improve the attention to transfer timing in EMaTO research, and adaptively select and transfer knowledge according to the optimization requirement of the population. RADE is applicable to address a wider range of real-world problems, where its adaptive knowledge transfer mechanism could significantly enhance optimization efficiency. Future work will pursue three interconnected directions to advance requirement-driven transfer optimization. First, we will reduce manual parameter dependency by introducing learning methods to dynamically adjust the parameter values. Second, the framework will be implemented in other EC algorithms, such as particle swarm optimization and ant colony optimization algorithms. Finally, we will validate the efficacy of RADE in complex real-world applications, which often involve challenges as noisy objective functions.

REFERENCES

- [1] Z. -H. Zhan, L. Shi, K. C. Tan, and J. Zhang, "A survey on evolutionary computation for complex continuous optimization," *Artif. Intell. Rev.*, vol. 55, no. 1, p. 59–110, Jan. 2022.
- [2] Z. -H. Zhan, J. Hong, J. -Y. Li, C. Wang, L. He, Z. Xu, and J. Zhang, "Artificial intelligence-based methods for protein structure prediction: A survey," *Artif. Intell. Rev.*, vol. 58, no. 328, 2025.
- [3] Y. Jiang, X. -X. Xu, M. -Y. Zheng, and Z. -H. Zhan, "Evolutionary computation for unmanned aerial vehicle path planning: A survey," *Artif. Intell. Rev.*, vol. 57, no. 267, 2024.
- [4] Y. Jiang, Z. -H. Zhan, K. C. Tan, and J. Zhang, "Knowledge learning for evolutionary computation," *IEEE Trans. Evol. Comput.*, vol. 29, no. 1, pp. 16–30, Feb. 2025.
- [5] V. Palakonda and J. -M. Kang, "Pre-DEMO: Preference-inspired differential evolution for multi/many-objective optimization," *IEEE Trans. Cybern.*, vol. 53, no. 12, pp. 7618–7630, Dec. 2023.
- [6] A. Gupta, Y. S. Ong, and L. Feng, "Insights on transfer optimization: because experience is the best teacher," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 2, no. 1, pp. 51–64, Feb. 2018.
- [7] T. Wei, S. Wang, J. Zhong, D. Liu, and J. Zhang, "A review on evolutionary multitask optimization: trends and challenges," *IEEE Trans. Evol. Comput.*, vol. 26, no. 5, pp. 941–960, Oct. 2022.
- [8] H. Zhao, X. -H. Ning, X. -T. Liu, C. Wang, and J. Liu, "What makes evolutionary multi-task optimization better: A comprehensive survey," *Appl. Soft Comput.*, vol. 145, pp. 110545, Sep. 2023.
- [9] X. -Y. Wang, Q. -T. Yang, Y. Jiang, K. C. Tan, J. Zhang, and Z. -H. Zhan, "Fine-grain knowledge transfer-based multitask particle swarm optimization with dual clustering-based task generation for high-dimensional feature selection," in *Proc. Genet. Evol. Comput. Conf.*, 2024, pp. 1506–1514.
- [10] Y. Wu *et al.*, "Evolutionary multimodal optimization with two-stage bi-directional knowledge transfer strategy for point cloud registration," *IEEE Trans. Evol. Comput.*, vol. 28, no. 1, pp. 62–76, Feb. 2024.
- [11] Y. Hou, X. -Y. Guo, H. -G. Han, and J. -J. Wang, "Knowledge-driven ant colony optimization algorithm for vehicle routing problem in instant delivery peak period," *Appl. Soft Comput.*, vol. 145, p. 110551, Sep. 2023.
- [12] C. Lyu, Y. Shi, L. Sun, and C.-T. Lin, "Community detection in multiplex networks based on evolutionary multitask optimization and evolutionary clustering ensemble," *IEEE Trans. Evol. Comput.*, vol. 27, no. 3, pp. 728–742, 2023.
- [13] B. H. T. Thanh, L. V. Cuong, T. B. Thang, and N. H. Long, "Ensemble multifactorial evolution with biased skill-factor inheritance for many-task optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 6, pp. 1735–1749, Dec. 2023.
- [14] S. -H. Wu, J. -Y. Li, Z. -H. Zhan, and J. Zhang, "Multi-view transfer with marginal and conditional alignment for many-task optimization," in *Proc. IEEE Congr. Evol. Comput.*, 2023, pp. 1–8.
- [15] N. Kumar, A. Narang, and B. Lall, "Kullback-leibler divergence-based regularized normalization for low-resource tasks," *IEEE Trans. Artif. Intell.*, vol. 5, pp. 2638–2650, Jun. 2024.
- [16] L. Zhou, L. Feng, J. Zhong, Z. Zhu, B. Da, and Z. Wu, "A study of similarity measure between tasks for multifactorial evolutionary algorithm," in *Proc. ACM Genet. Evol. Comput. Conf.*, 2018, pp. 229–230.
- [17] L. Feng *et al.*, "Evolutionary multitasking via explicit autoencoding," *IEEE Trans. Cybern.*, vol. 49, no. 9, pp. 3457–3470, Sept. 2019.
- [18] S. H. Wu, Z. H. Zhan, K. C. Tan, and J. Zhang, "Orthogonal transfer for multitask optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 1, pp. 185–200, Feb. 2023.
- [19] S. Li, W. Gong, L. Wang, and Q. Gu, "Evolutionary multitasking via reinforcement learning," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 1, pp. 762–775, Feb. 2024.
- [20] H. Xu, A. K. Qin, and S. Xia, "Evolutionary multitask optimization with adaptive knowledge transfer," *IEEE Trans. Evol. Comput.*, vol. 26, no. 2, pp. 290–303, Apr. 2022.
- [21] D. Yazdani *et al.*, "Robust optimization over time: A critical review," *IEEE Trans. Evol. Comput.*, vol. 28, no. 5, pp. 1265–1285, Oct. 2024.
- [22] L. Zhou *et al.*, "Toward adaptive knowledge transfer in multifactorial evolutionary computation," *IEEE Trans. Cybern.*, vol. 51, no. 5, pp. 2563–2576, May 2021.
- [23] X. Xue *et al.*, "Affine transformation-enhanced multifactorial optimization for heterogeneous problems," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 6217–6231, July 2022.
- [24] A. Gupta, Y. -S. Ong, and L. Feng, "Multifactorial evolution: toward evolutionary multitasking," *IEEE Trans. Evol. Comput.*, vol. 20, no. 3, pp. 343–357, Jun. 2016.
- [25] L. Feng *et al.*, "An empirical study of multifactorial PSO and multifactorial DE," in *Proc. IEEE Congr. Evol. Comput.*, 2017, pp. 921–928.
- [26] Z. -H. Zhan, J. Y. Li, S. Kwong, and J. Zhang, "Learning-aided evolution for optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 6, pp. 1794–1808, Dec. 2023.
- [27] X. F. Liu, Z. -H. Zhan, Y. Gao, J. Zhang, S. Kwong, and J. Zhang, "Coevolutionary particle swarm optimization with bottleneck objective learning strategy for many-objective optimization," *IEEE Trans. Evol. Comput.*, vol. 23, no. 4, pp. 587–602, Aug. 2019.
- [28] J. Y. Li, K. J. Du, Z. -H. Zhan, H. Wang, and J. Zhang, "Distributed differential evolution with adaptive resource allocation," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 2791–2804, May 2023.
- [29] J. Hong, Z. -H. Zhan, L. He, Z. Xu, and J. Zhang, "Protein structure prediction using a new optimization-based evolutionary and explainable artificial intelligence approach," *IEEE Trans. Evol. Comput.*, vol. 29, no. 3, pp. 646–660, June. 2025.
- [30] K. K. Bali, Y. -S. Ong, A. Gupta, and P. S. Tan, "Multifactorial evolutionary algorithm with online transfer parameter estimation: MFEA-II," *IEEE Trans. Evol. Comput.*, vol. 24, no. 1, pp. 69–83, Feb. 2020.
- [31] Z. Tang, M. Gong, Y. Wu, W. Liu, and Y. Xie, "Regularized evolutionary multitask optimization: Learning to inter-task transfer in aligned subspace," *IEEE Trans. Evol. Comput.*, vol. 25, no. 2, pp. 262–276, Apr. 2021.

> REPLACE THIS LINE WITH YOUR MANUSCRIPT ID NUMBER (DOUBLE-CLICK HERE TO EDIT) <

- [32]G. Li, Q. Zhang, and Z. Wang, "Evolutionary competitive multitasking optimization," *IEEE Trans. Evol. Comput.*, vol. 26, no. 2, pp. 278-289, Apr. 2022.
- [33]G. Li, Q. Lin, and W. Gao, "Multifactorial optimization via explicit multipopulation evolutionary framework," *Inform. Sci.*, vol. 512, pp. 1555-1570, Feb. 2020.
- [34]J.-Y. Li, Z.-H. Zhan, K. C. Tan, and J. Zhang, "A meta-knowledge transfer-based differential evolution for multitask optimization," *IEEE Trans. Evol. Comput.*, vol. 26, no. 4, pp. 719-734, Aug. 2022.
- [35]C. Wang, J. Liu, K. Wu, and Z. Wu, "Solving multitask optimization problems with adaptive knowledge transfer via anomaly detection," *IEEE Trans. Evol. Comput.*, vol. 26, no. 2, pp. 304-318, Apr. 2022.
- [36]Y. Jiang, Z.-H. Zhan, K. C. Tan, and J. Zhang, "Block-level knowledge transfer for evolutionary multitask optimization," *IEEE Trans. Cybern.*, vol. 54, no. 1, pp. 558-571, Jan. 2024.
- [37]T. Zhang, W. Gong, and Y. Li, "Multitask differential evolution with adaptive dual knowledge transfer," *Appl. Soft Comput.*, vol. 165, p.112040, 2024.
- [38]Y. Jiang, Z.-H. Zhan, K. C. Tan, and J. Zhang, "A bi-objective knowledge transfer framework for evolutionary many-task optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 5, pp. 1514-1528, Oct. 2023.
- [39]Y. Hou, Y. Shen, H. Han, and J. Wang, "Many-task differential evolutionary algorithm based on bi-space similarity," *IEEE Trans. Evol. Comput.*, vol. 29, no. 4, pp. 1215-1226, Aug. 2025.
- [40]H. Han, X. Bai, H. Han, Y. Hou, and J. Qiao, "Self-adjusting multitask particle swarm optimization," *IEEE Trans. Evol. Comput.*, vol. 26, no. 1, pp. 145-158, Feb. 2022.
- [41]Y. -T. Dai, X. -F. Liu, Y. Fang, Z. -H. Zhan and J. Zhang, "Multi-source knowledge fusion based on graph attention networks for many-Task Optimization," *IEEE Trans. Evol. Comput.*, early access, Sep. 23, 2024, doi: 10.1109/TEVC.2024.3465542.
- [42]S. -H. Wu, Z. -H. Zhan, K. C. Tan, and J. Zhang, "Transferable adaptive differential evolution for many-task optimization," *IEEE Trans. Cybern.*, vol. 53, no. 11, pp. 7295-7308, Nov. 2023.
- [43]Y. Li, W. Gong, and S. Li, "Multitasking optimization via an adaptive solver multitasking evolutionary framework," *Inform. Sci.*, vol. 630, pp. 688-712, Jun. 2023.
- [44]Y. Jiang, Z. -H. Zhan, K. C. Tan, S. Kwong, and J. Zhang, "Knowledge structure preserving-based evolutionary many-task optimization," *IEEE Trans. Evol. Comput.*, vol. 29, no. 2, pp. 287-301, April 2025.
- [45]X. Wu, W. Wang, H. Yang, H. Han, and J. Qiao, "Multitasking feedback optimization algorithm based on an evolutionary state estimator," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 3, pp. 2554-2569, June 2024.
- [46]Z.-G. Chen, Z. -H. Zhan, S. Kwong, and J. Zhang, "Evolutionary computation for intelligent transportation in smart cities: A survey," *IEEE Comput. Intell. Mag.*, vol. 17, no. 2, pp. 83-102, May, 2022.
- [47]G. -Y. Lin et al., "A landscape-aware differential evolution for multimodal optimization problems," *IEEE Trans. Evol. Comput.*, early access, Feb. 2025, doi: 10.1109/TEVC.2025.3545602.
- [48]Y. Chen, J. Zhong, L. Feng, and J. Zhang, "An adaptive archive-based evolutionary framework for many-task optimization," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 4, no. 3, pp. 369-384, Jun. 2020.
- [49]T. J. Choi and C. W. Ahn, "An improved LSHADE-RSP algorithm with the Cauchy perturbation: iLSHADE-RSP," *Knowl.-Based Syst.*, vol. 215, Mar. 2021.
- [50]L. Feng, K. Qin, A. Gupta, Y. Yuan, Y.-S. Ong, and X. Chi, "IEEE CEC 2019 competition on evolutionary multitask optimization," http://www.bdsc.site/websites/MTO_competition_2019/MTO_Competition_CEC_2019.html, 2019.
- [51]L. Feng et al., "IEEE CEC 2022 competition on evolutionary multitask optimization," http://www.bdsc.site/websites/MTO_competition_2021/MTO_Competition_WCCI_2022.html, 2022.
- [52]B. Da et al., "Evolutionary multitasking for single-objective continuous optimization: benchmark problems, performance metric, and baseline results," arXiv preprint arXiv:1706.03470, Jun. 2017.
- [53]Y. Yuan et al., "Evolutionary multitasking for multiobjective continuous optimization: benchmark problems, performance metrics and baseline results," arXiv preprint arXiv:1706.02766, Jun. 2017.
- [54]J. Carrasco, S. Garcia, M. M. Rueda, S. Das, and F. Herrera, "Recent trends in the use of statistical tests for comparing swarm and

evolutionary computing algorithms: practical guidelines and a critical review," *Swarm Evol. Comput.*, vol. 54, May 2020.

- [55]W. Gao and Y. Li, "Solving a new test set of nonlinear equation systems by evolutionary algorithm," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 406-415, Jan. 2023.



Zhi-Fan Tang (Student Member, IEEE) received the B.S. degree in computer science and technology from South China University of Technology, Guangzhou, China, in 2022, where she is currently pursuing the Ph.D. degree in computer science and technology with the School of Computer Science and Engineering.

Her research interests include multi-task optimization, large-scale optimization, evolutionary algorithms, and their applications in real-world.



Zhi-Hui Zhan (Fellow, IEEE) received the bachelor's and Ph.D. degrees in computer science from Sun Yat-Sen University, Guangzhou China, in 2007 and 2013, respectively.

He is currently a Changjiang Scholar Young Professor and Gifted Professor with the College of Artificial Intelligence, Nankai University, Tianjin, China. His current research interests include evolutionary computation, swarm intelligence, and their applications

in real-world problems.

Prof. Zhan is an IEEE Fellow. He was a recipient of the IEEE Computational Intelligence Society (CIS) Outstanding Early Career Award in 2021, the Outstanding Youth Science Foundation from National Natural Science Foundations of China (NSFC) in 2018, and the Wu Wen-Jun Artificial Intelligence Excellent Youth from the Chinese Association for Artificial Intelligence in 2017. He is listed as the Highly Cited Researcher by Clarivate Analytics, is listed as the World's Top 2% Scientist for both Career-Long Impact and Year Impact in Artificial Intelligence, and is listed as the Elsevier Highly Cited Chinese Researcher in Computer Science since 2014. He is currently an Associate Editor of the *IEEE Transactions on Evolutionary Computation*, the *IEEE Transactions on Emerging Topics in Computational Intelligence*, the *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, and the *IEEE Transactions on Artificial Intelligence*.



Yun Li (Fellow, IEEE) received the B.S. degree from Sichuan University, Chengdu, China, in 1984, the M.E. degree from the University of Electronic Science and Technology of China (UESTC), Chengdu, in 1987, and the Ph.D. degree from the University of Strathclyde, Glasgow, U.K., in 1990.

He is currently a Changjiang Chair Professor with the Shenzhen Institute for Advanced Study, UESTC.

He has published 300 papers, and one of them since publication in the *IEEE Transactions on Control System Technology* in 2005 has been its most popular article almost every month. He is interested in the next generation, explainable artificial intelligence (AI) and has supervised over 30 Ph.D. students focusing on computational AI and its engineering/industrial applications since 1991. Prof. Li is a U.K. Chartered Engineer.



Jun Zhang (Fellow, IEEE) obtained his PhD degree in Electrical Engineering from the City University of Hong Kong in 2002.

His primary research interests lie in the field of Computational Intelligence (CI), with a strong focus on Evolutionary Computation (EC) and its wide-ranging applications. Professor Zhang has authored more than 600 peer-reviewed publications, including over 230 papers in IEEE Transactions. He has been recognized as a Clarivate Highly Cited Researcher, ranking within the top 1% worldwide in the field of Computer Science. In 2011, he was awarded the Outstanding Young Scientist Fund by the National Natural Science Foundation of China (NSFC).

He currently serves as an Associate Editor of the *IEEE Transactions on Artificial Intelligence* and the *IEEE Transactions on Cybernetics*.